

ACT Prep · Week 1 · Roots & Formulas

Ten word roots to grow your vocabulary, and every formula from this week's lesson.

Word roots

Roots are the building blocks of words. When you meet a word you don't know on the ACT, break it into parts: the root usually gives you enough of the meaning to answer. Your list grows every week; next week's sheet adds new roots and reviews these.

com- / con- TOGETHER, WITH

- **composite**: made of parts put together
- **consensus**: general agreement
- **converge**: to come together

Your ACT **composite** puts your English, Math and Reading scores together.

pos / pon PUT, PLACE

- **component**: one part of a whole
- **postpone**: to put off until later
- **position**: the place something is put

"The meeting was **postponed**" already means it was moved later; don't say it twice.

re- BACK, AGAIN

- **redundant**: repeating what was already said; more than needed
- **revise**: to look at again and improve
- **reject**: to throw back; refuse

On ACT English, cut **redundant** words: the shortest correct choice usually wins.

dict SAY, SPEAK

- **contradict**: to say the opposite
- **predict**: to say before it happens
- **verdict**: a judgment that is spoken

A wrong Reading answer often **contradicts** a line in the passage.

fer CARRY, BRING

- **infer**: to conclude from evidence
- **transfer**: to carry across
- **reference**: something you carry back to or point to

An **inference** question asks what the passage suggests, not what it says word for word.

cis / cid CUT

- **concise**: cut down to only what is needed
- **precise**: exact; cut to the right size
- **incision**: a cut

The ACT rewards **concise** writing: say it once, clearly.

vid / vis SEE

- **evidence**: what can be seen that proves something
- **visible**: able to be seen
- **revise**: to see again

Before you pick a Reading answer, point to the **evidence** in the passage.

graph WRITE

- **paragraph**: a written section of text
- **graph**: a written picture of data
- **biography**: a written account of a life

ACT Reading passages number each **paragraph** so questions can point to it.

pre- BEFORE

- **predict**: to say before
- **precede**: to come before
- **preview**: to see before

Skim the questions to **preview** what the passage will ask about.

in- / im- / ir- NOT

- **inaccurate**: not correct
- **implausible**: not believable
- **irrelevant**: not related to the topic

If a sentence is **irrelevant** to the paragraph, the ACT wants it deleted.

Five minutes a day: cover the words and say what each root means, then name one word that uses it.

Formulas from this week

The ACT gives you no formula sheet, so these need to live in your head. Say each one out loud, then check yourself with the example.

FORMULA	RULE	EXAMPLE
ACT composite	$(\text{English} + \text{Math} + \text{Reading}) \div 3$, round to nearest whole (.5 rounds up)	$22 + 18 + 21 = 61$; $61 \div 3 = 20.33 \rightarrow 20$
Seconds per question	$(\text{section minutes} \times 60) \div \text{number of questions}$	English: $35 \times 60 \div 50 = 42$ s
Percent of a number	percent as a decimal $\times$ the number	15% of 60 = $0.15 \times 60 = 9$
Percent change	$\text{new} = \text{old} \times (1 \pm \text{rate})$	Up 25% on \$40: $40 \times 1.25 = \$50$
Discount, then tax	$\text{price} \times (1 - \text{discount})$, then $\times (1 + \text{tax})$	\$80, 25% off, 6% tax: $80 \times 0.75 = 60$; $60 \times 1.06 = \$63.60$
Average (mean)	$\text{sum} \div \text{number of items}$	80, 90, 85 $\rightarrow 255 \div 3 = 85$
Distance	$\text{distance} = \text{rate} \times \text{time}$	55 mph for 3 h = 165 miles
Slope	$(y_2 - y_1) \div (x_2 - x_1)$	(2, 3) to (4, 11): $8 \div 2 = 4$
Line equation	$y = mx + b$ (m = slope, b = y-intercept)	$y = 2x + 5$: slope 2, crosses at 5
Triangle area	$\frac{1}{2} \times \text{base} \times \text{height}$	base 10, height 6 $\rightarrow 30$
Circle area	$A = \pi r^2$	$r = 3 \rightarrow 9\pi \approx 28.3$
Circumference	$C = 2\pi r$	$r = 3 \rightarrow 6\pi \approx 18.8$
Pythagorean theorem	$a^2 + b^2 = c^2$	legs 6 and 8 $\rightarrow$ hypotenuse 10
Probability	$\text{favorable outcomes} \div \text{total outcomes}$	3 red of 12 marbles $\rightarrow 3/12 = 1/4$